

A/B Test Experiment and Recommendation Report: Closed-Domain RAG System

1. Experiment Specification

Hypothesis This experiment was designed to evaluate whether an improved retrieval configuration (Variant B) delivers meaningfully better answer quality compared to the current baseline configuration (Variant A).

The null hypothesis assumed no difference in performance between the two variants. The alternative hypothesis posited that Variant B would improve correctness and groundedness while reducing hallucinations.

Success Metrics The primary metrics evaluated included:

- Correctness Rate
- Groundedness Rate
- Hallucination Rate
- Latency (Mean and P95)
- Refusal Accuracy
- Business KPI: Trusted Answer Rate

Randomization Method Queries were randomly assigned to either Variant A or Variant B using a 50/50 split. This ensured an unbiased comparison and equal exposure for both configurations across the test set.

Sample Size and Duration A statistical power analysis determined that a minimum of 218 samples per variant was required. The experiment exceeded this threshold, using 250 samples per variant. This provided additional statistical confidence in the results.

2. Experiment Results

Variant A vs. Variant B Performance

- **Correctness:** 79.2% vs. 88.0%
- **Groundedness:** 83.2% vs. 92.0%
- **Hallucination Rate:** 8.4% vs. 3.6%
- **Mean Latency:** 2.10s vs. 2.58s
- **P95 Latency:** 2.77s vs. 3.22s

Statistical Evaluation

- Correctness p-value = 0.00788 (statistically significant)
- Groundedness p-value = 0.00283 (statistically significant)
- Latency p-value $\approx$ 0.000 (statistically significant)

The results show that Variant B delivered clear and statistically significant improvements across all quality metrics, with a notable reduction in hallucinations. As expected, this came with an increase in latency.

3. Interpretation

Variant B demonstrated a meaningful step forward in answer quality. Correctness improved by nearly 9 percentage points, groundedness increased by almost 9 points, and the hallucination rate was reduced by more than 57%. These gains significantly enhance the reliability and trustworthiness of the RAG system's outputs.

The trade-off is an increase in response time — mean latency rose by approximately 0.48 seconds, with a similar increase at the P95 level. While this increase is noticeable, the substantial quality improvements appear to justify the modest latency cost in most use cases.

4. Recommendation

I recommend proceeding with the deployment of **Variant B** into production.

The significant improvements in correctness, groundedness, and the sharp reduction in hallucinations strongly outweigh the increase in latency. These enhancements should translate into higher user trust and better overall system performance.

That said, latency should be closely monitored after launch. Where feasible, further optimization efforts should be prioritized to reduce the performance gap without compromising the quality gains achieved.